

Portfolio Project: Telco Extra

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Organizations in the telecommunications industry increasingly depend on analytic methods to better understand customer behavior, anticipate attrition, and inform strategic decision-making. The Telco Extra dataset offers a structured view of customer demographics, service attributes, and outcomes such as churn and income. Because customer income in this dataset is obtained from an external vendor, the company seeks to determine whether internal data can be used to reliably model income and reduce dependence on third-party sources. At the same time, understanding the drivers of customer churn is critical for improving retention and long-term profitability.

The purpose of this study is to examine the characteristics of Telco Extra customers and to evaluate the relationships between key customer attributes, churn, and income. Descriptive statistics are first used to establish a comprehensive profile of the customer sample across nine variables. A binary logistic regression model is then employed to identify factors associated with customer churn. Finally, a multiple linear regression model is developed to assess how demographic and service-related variables relate to customer income.

Collectively, these analyses provide an empirical foundation for understanding how stability, demographic characteristics, and customer segmentation influence both retention and financial outcomes. By integrating descriptive and inferential methods, this paper demonstrates how internal customer data can be leveraged to generate actionable insight, supporting efforts to improve retention strategies and evaluate the feasibility of predicting income without reliance on external data sources.

Sample Characteristics

Descriptive data analytics focuses on summarizing and describing key features of a dataset in order to provide insight and understanding prior to more advanced analysis (zyBooks, 2022). In this study, descriptive statistics are used to examine the characteristics of Telco Extra customers across demographic, geographic, and service-related variables. By presenting measures of central tendency, dispersion, and categorical distributions, this section establishes a clear baseline profile of the customer sample.

In applied analytics contexts, descriptive statistics are commonly used to set benchmarks and support interpretation of subsequent analytical models. For example, within Six Sigma and business analytics frameworks, descriptive measures help identify patterns, variation, and overall data structure before moving into predictive or inferential techniques (Chau, 2025). Accordingly, the following section summarizes the sample characteristics for nine key variables using appropriate statistical measures and SAS-generated visualizations.

Age

Based on the results of descriptive statistical analysis (Table 1), the age variable in the Telco Extra dataset includes 1,000 observations, with customer ages ranging from 18 to 77 years. The mean age of customers is 41.68 years ($SD = 12.56$), while the median age is 40 years. This indicates that the distribution is centered around early middle adulthood. The interquartile range spans from 32 to 51 years, suggesting that the middle 50% of customers fall within a relatively narrow working-age band. These descriptive measures support the visual interpretation of the histogram and provide context for the observed concentration of customers in early adulthood through late middle age.

Table 1

Descriptive Statistics for Customer Age

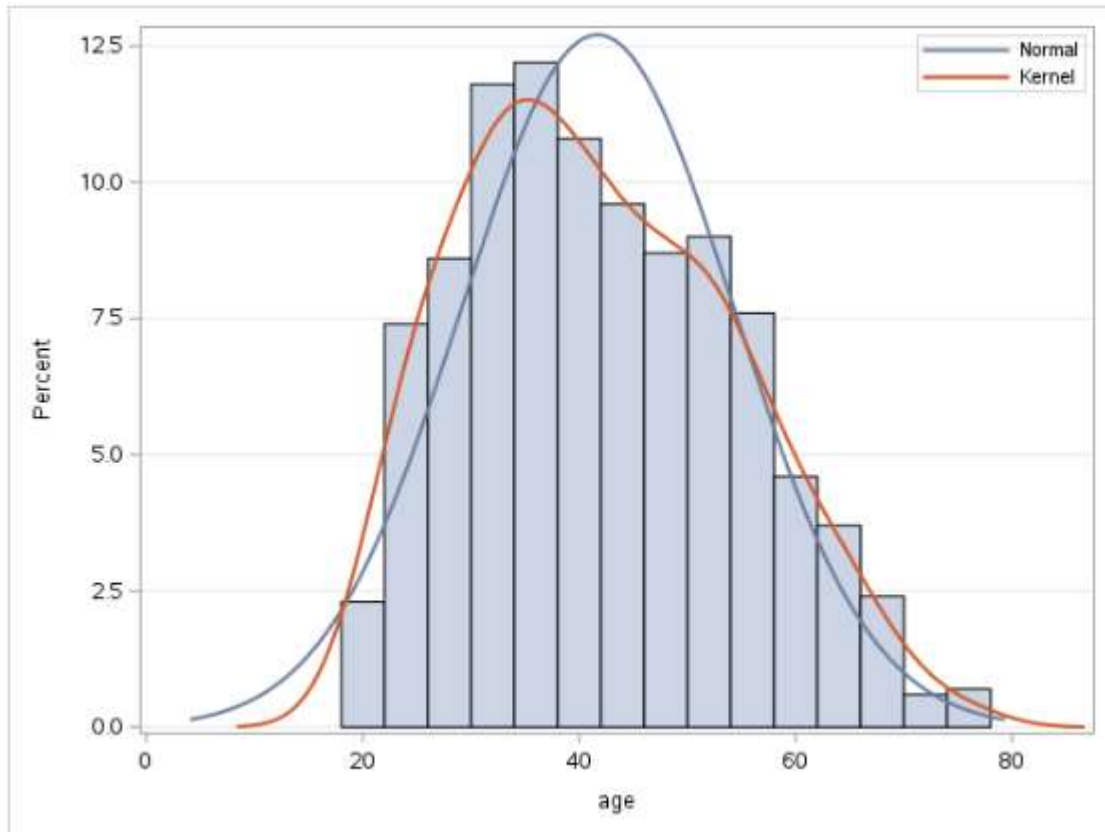
Analysis Variable : age age							
N	Mean	Std Dev	Median	Minimum	Maximum	Lower Quartile	Upper Quartile
1000	41.6840000	12.5588163	40.0000000	18.0000000	77.0000000	32.0000000	51.0000000

Note. This table summarizes descriptive statistics for customer age in the Telco Extra dataset.

Figure 1 is an age histogram for customers at Telco Extra. It indicates that the minimum observed customer age is approximately 20 years old, which is consistent with eligibility requirements for telecommunications services. Individuals younger than this age are typically unable to independently enter into service contracts. The distribution closely follows a normal curve which suggests a relatively balanced customer-age profile. However, there is a slight right skew indicating a longer tail extending toward older age groups. The core customer base for the company falls between the ages of 25 and 59. This range is likely to reflect working-age individuals who have consistent income and a greater need for telecommunications services for both personal and professional use. As age increases beyond this range, the number of customers begins to decline. This may be attributed to reduced technology adoption, lower service usage needs (e.g. no professional usage), or a preference for alternative companies among older populations. The slight right skew suggests that while older customers remain part of the customer base, they represent a smaller proportion compared to middle-aged customers. Overall, the age distribution supports the conclusion that the company's services primarily appeal to adults in early adulthood through late middle age.

Figure 1

Distribution of Customer Age



Note. The histogram displays the distribution of customer age in the dataset. The bars represent the percentage of customers within each age range, while the overlaid curves illustrate the normal distribution and kernel density estimation used to visualize the overall shape and spread of the data.

Years at Current Address

The descriptive statistics displayed in Table 2 indicates that customers have lived at their current address anywhere from 0 to 55 years. The mean length of residence is 11.55 years ($SD = 10.89$), while the median is 9 years, suggesting that more than half of customers have lived at their current address for less than a decade. The interquartile range extends from 3 to 18 years, indicating substantial variability in residential tenure across the sample. These summary statistics support the visual evidence of a highly right-skewed distribution, where a small number of long-term residents increase the mean relative to the median.

Table 2

Descriptive Statistics for Years at Current Address

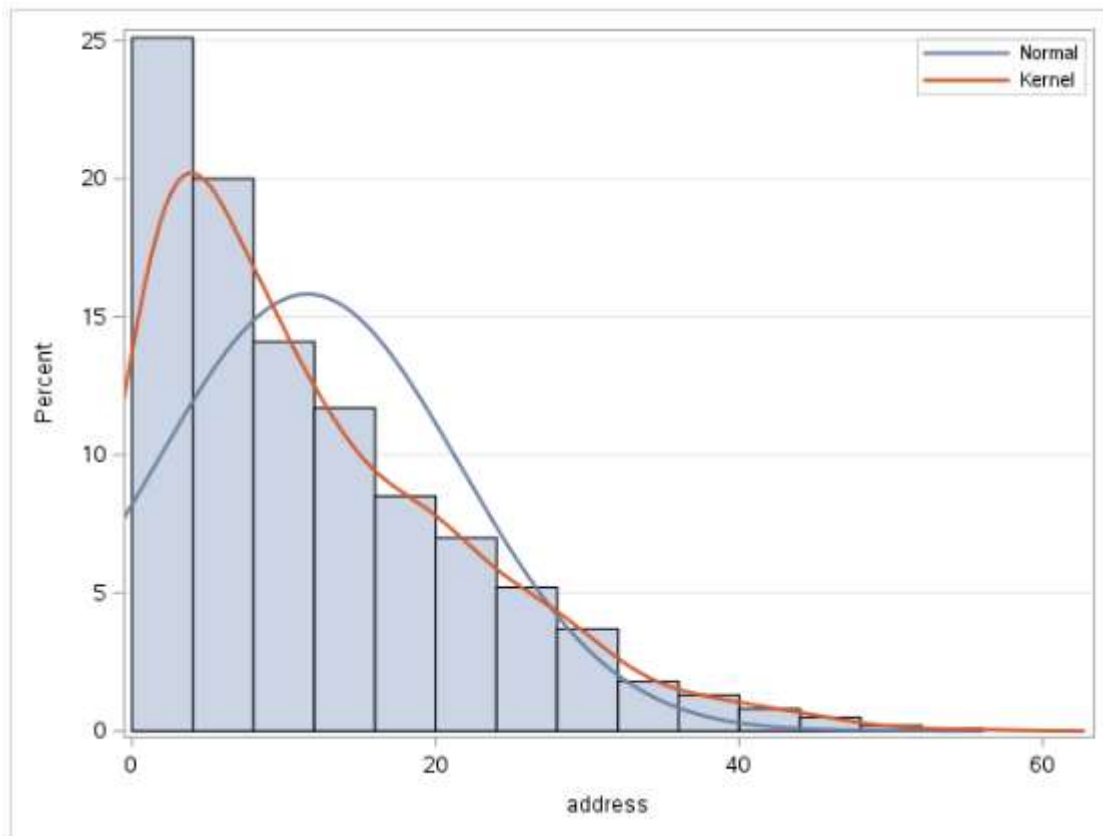
Analysis Variable : address address							
N	Mean	Std Dev	Median	Minimum	Maximum	Lower Quartile	Upper Quartile
1000	11.5510000	10.0886813	9.0000000	0	55.0000000	3.0000000	18.0000000

Note. This table summarizes descriptive statistics for years at current address in the Telco Extra dataset.

The histogram for years at current address (Figure 2) is heavily right skewed with most customers concentrated in the lower range of the distribution. A large share of customers (roughly 60% of the customer population) have lived at their current residence for fewer than 10 years, indicating relatively high residential mobility within the dataset. This pattern may reflect increased mobility among customers due to job changes, housing affordability, or lifestyle factors. As the number of years at the same address increases, the frequency of customers declines sharply, with only a small proportion remaining at one location for several decades. The long right tail highlights the presence of long-term residents, though they represent a minority of the customer base. The uneven spread and strong skewness of the distribution indicates that years at current address varies widely across customers rather than clustering around a central value.

Figure 2

Distribution of Years at Current Address



Note. This histogram illustrates the distribution of the number of years customers have lived at their current address. The bars represent the percentage of customers within each range of years, while the overlaid curves display the normal distribution and kernel density estimation used to assess the overall shape and spread of the data.

Gender

Gender in the Telco Extra dataset is recorded as a binary categorical variable coded as 0 and 1. The dataset does not provide labels indicating which value corresponds to a specific gender category; therefore, interpretation is limited to differences in representation rather than identification of male or female groups. Frequency results for gender are summarized in Table 3.

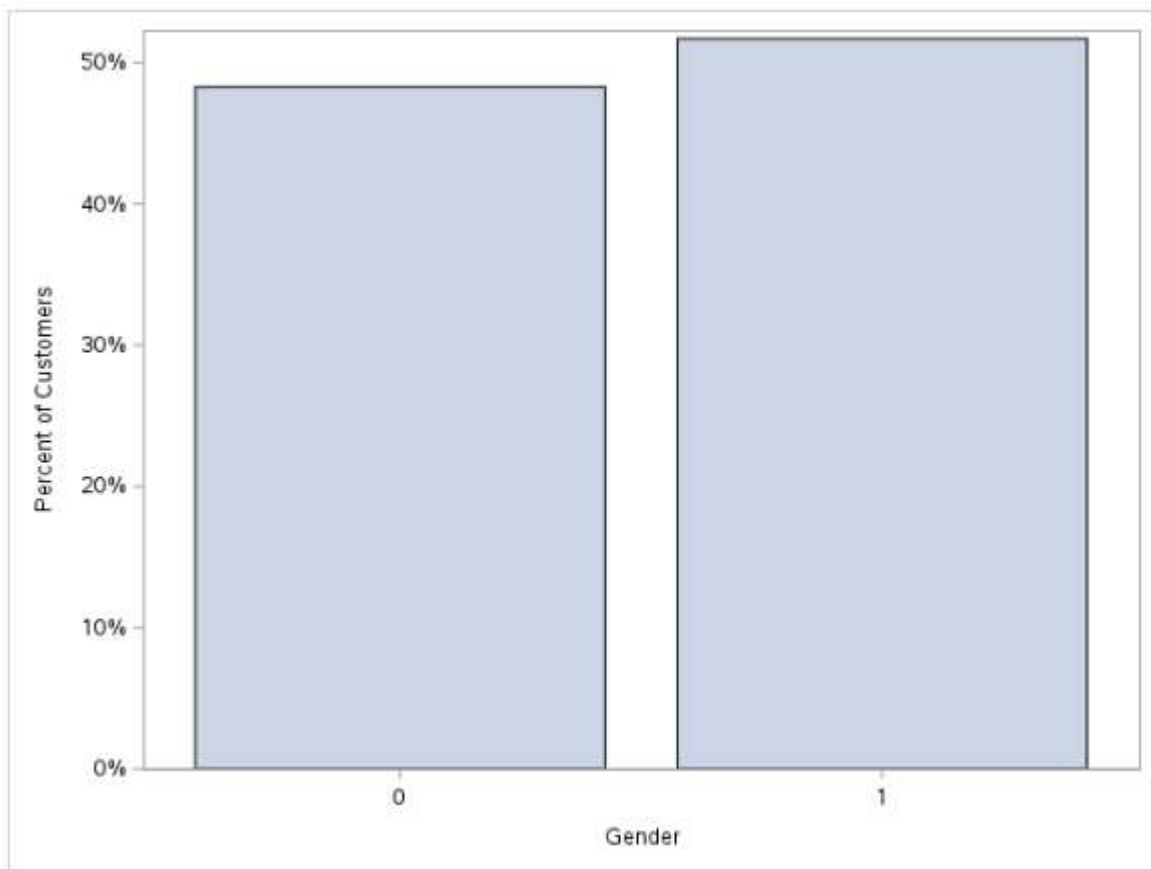
Table 3
Frequency Statistics for Customer Gender

gender		
gender	Frequency	Percent
0	483	48.30
1	517	51.70

Note. This table summarizes descriptive statistics for customer gender in the Telco Extra dataset.

Although the dataset does not specify which gender category corresponds to each coded value, the frequency distribution still provides useful insight into the overall demographic composition of the customer base. As shown in Figure 3, the sample is nearly evenly split between the two gender categories. This balanced distribution suggests that neither group is disproportionately represented in the dataset.

Figure 3
Frequency Chart for Customer Gender



Note. This chart displays the percentage distribution of customer genders in the Telco Extra dataset.

Level of Education

Table 4 is the distribution of education levels among Telco Extra customers. It reflects a range of educational background where higher values indicate higher levels of education.

Although the dataset does not include descriptive labels for each category, the frequency results provide insight into how customers are distributed across the education spectrum.

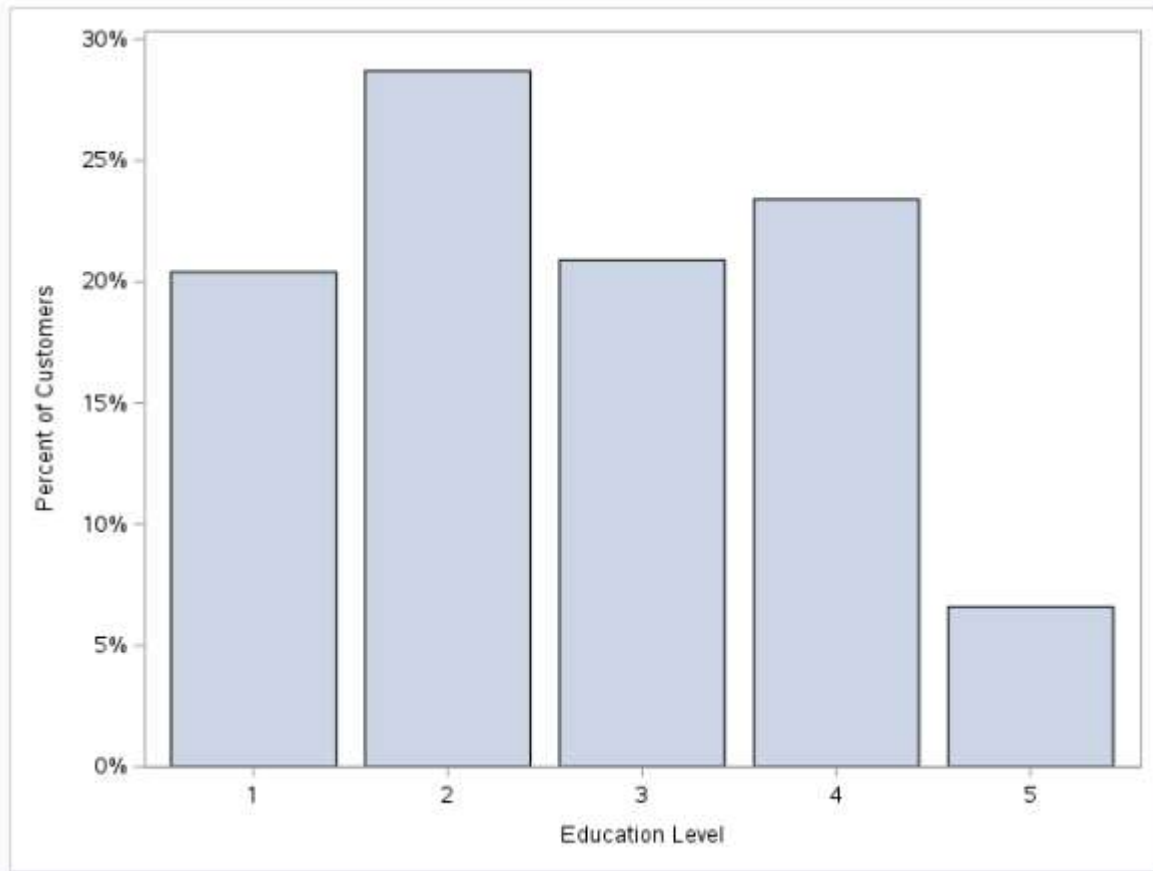
Table 4
Frequency Statistics for Level of Education

ed		
ed	Frequency	Percent
1	204	20.40
2	287	28.70
3	209	20.90
4	234	23.40
5	66	6.60

Note. This table summarizes descriptive statistics for education levels in the Telco Extra dataset.

Customers are most commonly represented at education level 2, which accounts for 28.7% of the sample. A substantial share of customers also falls into education levels 4 (23.4%), 3 (20.9%), and 1 (20.4%) which indicates that the majority of the customer base is concentrated within the lower to middle portions of the education scale. In contrast, education level 5 represents a much smaller segment of the population at 6.6%, suggesting that relatively few customers fall into the highest education category. The bar chart in Figure 4 highlights this pattern by showing a gradual decline in representation at the upper end of the education scale. This pattern may reflect broader population trends in education rather than a direct relationship between education level and customer participation as higher levels of education are generally less common in the population overall.

Figure 4
Frequency Chart for Level of Education



Note. This chart displays the percentage distribution of level of education in the Telco Extra dataset.

Income

Descriptive statistics for customer income (Table 5) indicate income values ranging from 9 to 1,688. The mean income is 77.54 (SD = 107.04) and the median income is 47. This indicates a substantial difference between the mean and median. This divergence reflects the influence of a small number of extremely high-income observations on the overall distribution. The interquartile range spans from 29 to 83, demonstrating that the middle 50% of customers are concentrated within a relatively narrow income band compared to the full range of observed values.

Table 5

Descriptive Statistics for Customer Income

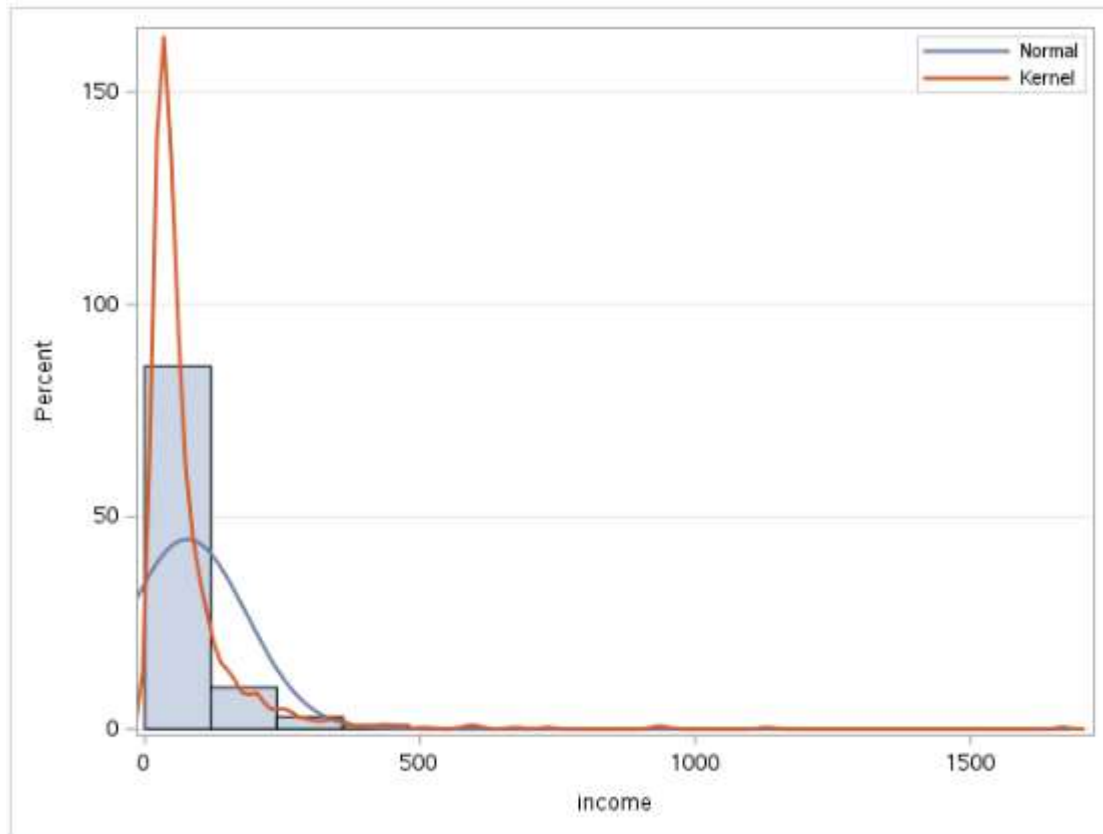
Analysis Variable : income income							
N	Mean	Std Dev	Median	Minimum	Maximum	Lower Quartile	Upper Quartile
1000	77.5350000	107.0441648	47.0000000	9.0000000	1668.00	29.0000000	83.0000000

Note. This table summarizes descriptive statistics for customer income in the Telco Extra dataset.

The income histogram (Figure 5) confirms this as it is highly right skewed. This is primarily due to the presence of several extremely high-income outliers. The majority of customers are concentrated at the lower end of the income scale, with approximately 85.4% of the population reporting income values below 60. An additional 9.8% of customers fall within the range of 60 to 180, while only a very small proportion of observations extend far beyond this range, creating a long right tail. These outliers significantly affect the overall shape of the distribution and cause a substantial deviation from a normal curve. The dataset does not provide identifying information regarding the unit or scale of measurement for income, making it unclear whether the values represent thousands of dollars, annual income in another currency, or an alternative standardized metric. As a result, interpretation is limited to relative comparisons rather than absolute monetary conclusions.

Figure 5

Distribution of Customer Income



Note. This histogram displays the distribution of customer income levels in the dataset. The bars represent the percentage of customers within each range of income, while the overlaid curves display the normal distribution and kernel density estimation used to assess the overall shape and spread of the data.

Marital Status

Marital status in the Telco Extra dataset is recorded as a binary categorical variable representing two marital categories. Frequency results are summarized in Table 6, with a corresponding bar chart shown in Figure 6. The distribution indicates that 50.5% of customers are coded as 0, while 49.5% are coded as 1, reflecting an almost even split between the two marital status categories.

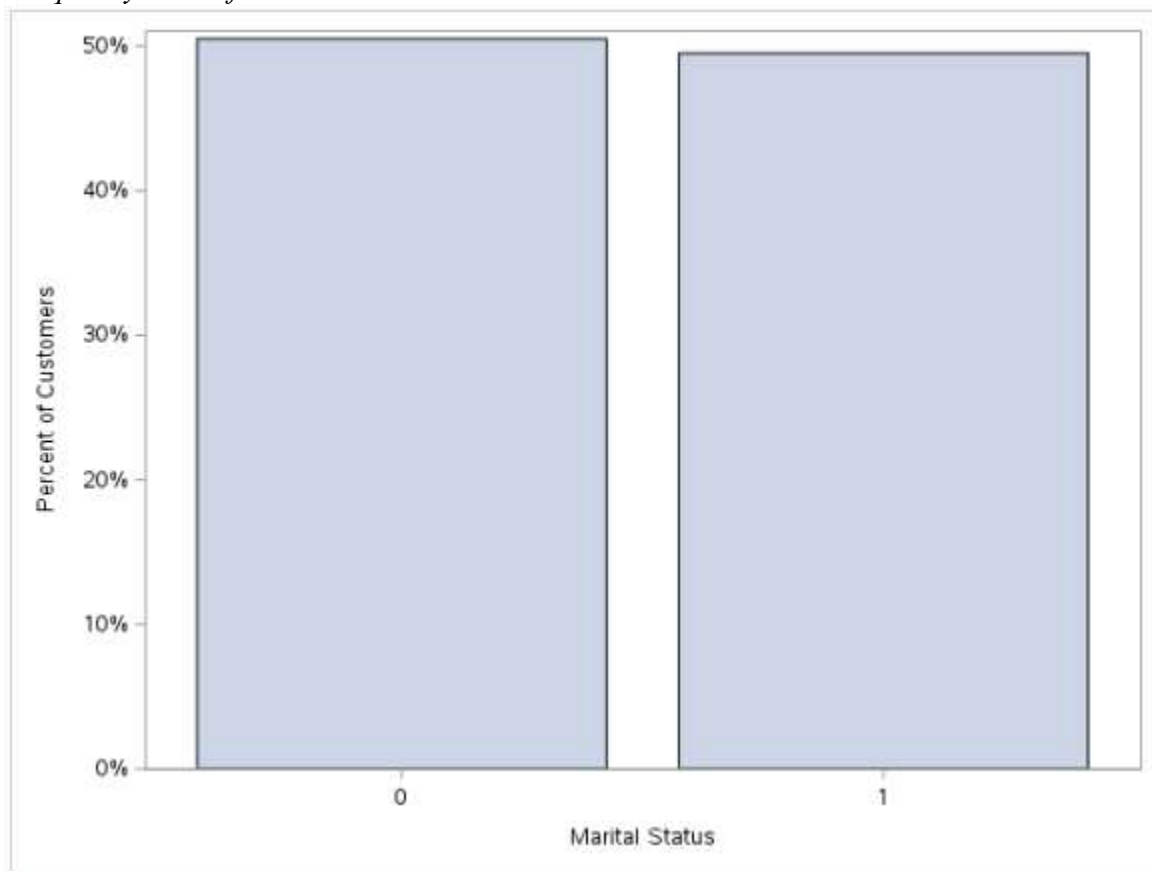
Table 6

Frequency Statistics for Customer Marital Status

marital				
marital	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	505	50.50	505	50.50
1	495	49.50	1000	100.00

Note. This table summarizes descriptive statistics for customer churn in the Telco Extra dataset. A value of 0 indicates customers who are not married and a value of 1 indicates customers who are married.

Figure 6
Frequency Chart for Customer Marital Status



Note. This chart displays the percentage of customers' marital status in the Telco Extra dataset.

This near balance aligns closely with broader population trends in the United States. Recent survey data indicate that approximately 53% of individuals between the ages of 25 and 54 are married, suggesting that the marital composition of the Telco Extra customer base is generally consistent with expectations for a working-age population (Enten, 2022). The

similarity between the sample distribution and national patterns suggests that marital status is not disproportionately skewed within the dataset.

It is also important to note that marital status is captured using only two categories, which does not account for alternative household arrangements such as cohabitation. Individuals who are living with a partner but are not legally married are therefore likely included in the non-married category. This may limit the granularity of insights related to relationship status.

Region

The regional distribution of customers in the Telco Extra dataset is relatively uniform across the three coded regions (1–3). As shown in Table 7 and Figure 7, 32.2% of customers are located in region 1, 33.4% in region 2, and 34.4% in region 3. The difference between the smallest and largest regional shares is minimal, with all regions falling within approximately two percentage points of one another.

Table 7

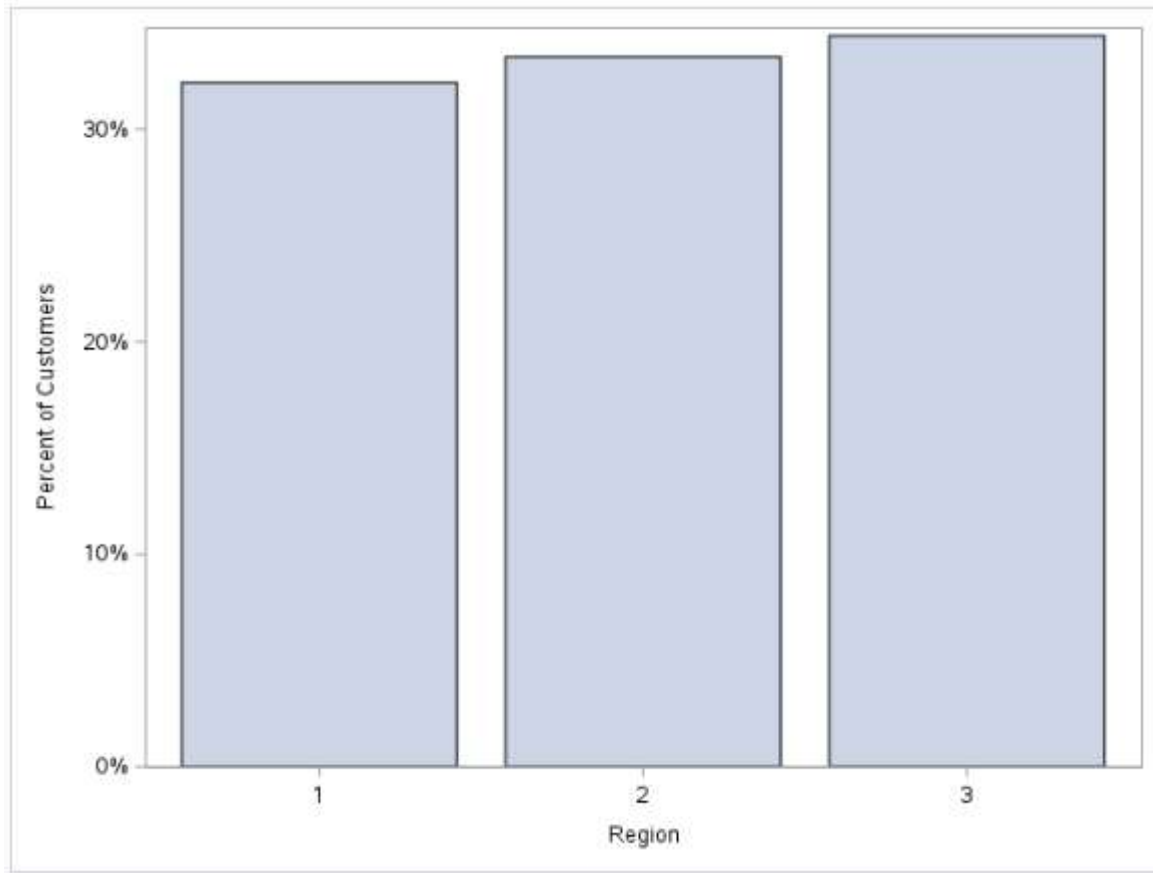
Frequency Statistics for Customer Regions

region		
region	Frequency	Percent
1	322	32.20
2	334	33.40
3	344	34.40

Note. This table summarizes descriptive statistics for customer regions in the Telco Extra dataset.

Figure 7

Frequency Chart for Customer Regions



Note. This chart displays the percentage of customers regions in the Telco Extra dataset.

This close alignment across regions suggests that the customer base is geographically well balanced with no single region accounting for a disproportionately large share of customers. As a result, regional representation is unlikely to introduce meaningful bias into subsequent analyses that include region as an explanatory variable.

Custcat

Customer category (custcat) in the Telco Extra dataset represents a categorical classification used to group customers into distinct segments based on shared characteristics or service-related attributes. Variables of this type are commonly used in customer analytics and machine learning applications to segment customers into meaningful groups, particularly in classification contexts such as the K-Nearest Neighbors (KNN) algorithm (Benli, 2023). In this

dataset, custcat is coded into four categories labeled 1 through 4, with each category representing a distinct customer group.

As shown in Table 8, customer categories 1 (28.6%) and 3 (28.1%) account for the largest share of the customer base, together representing more than half of the sample. Category 4 comprises 23.6% of customers, while smallest group is category 2 at 21.7%. The bar chart in Figure 8 illustrates that customer distribution across categories is relatively balanced, with no single category overwhelmingly dominant.

This distribution suggests that the dataset captures a diverse mix of customer segments rather than being concentrated in one primary category. Such balance is useful for subsequent analyses, as customer category variables are often included to capture differences in service usage or customer behavior across segments (Benli, 2023).

Table 8

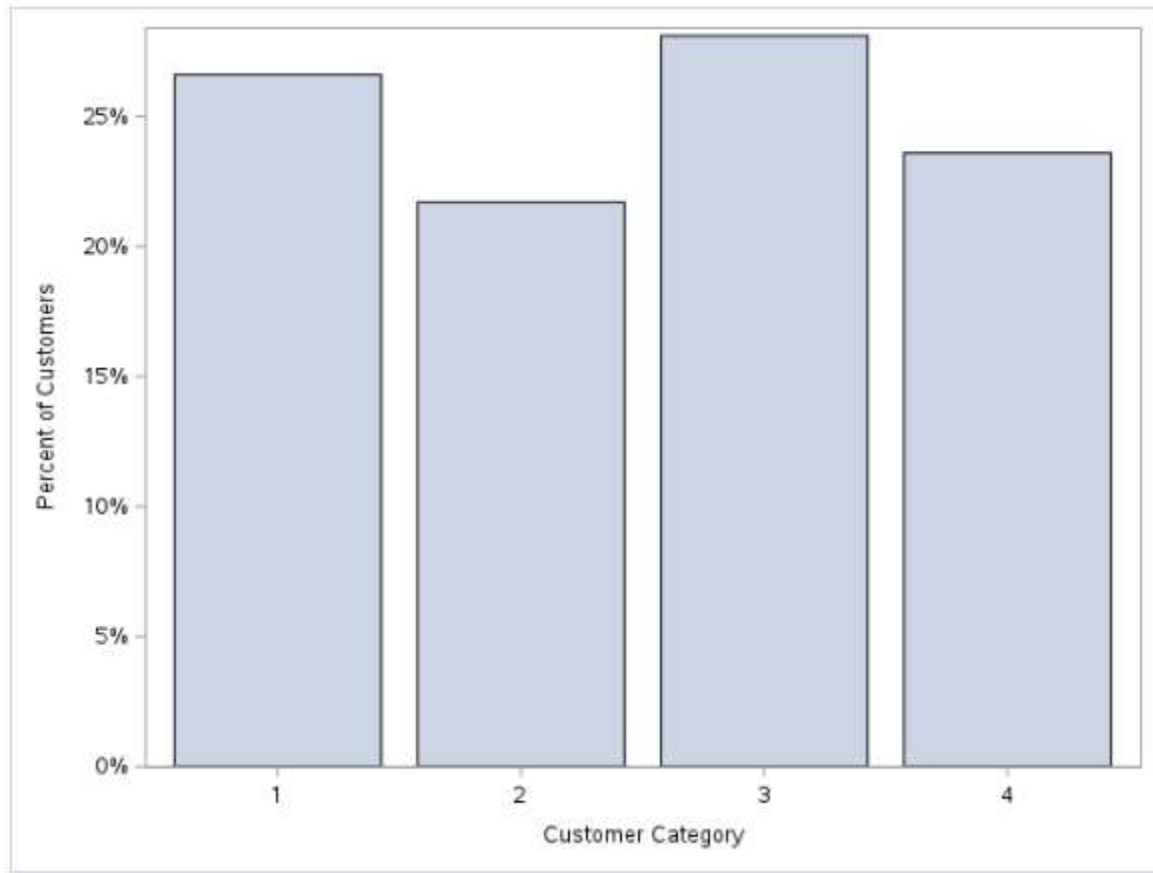
Frequency Statistics for Custcat

custcat		
custcat	Frequency	Percent
1	266	26.60
2	217	21.70
3	281	28.10
4	236	23.60

Note. This table summarizes descriptive statistics for customer churn in the Telco Extra dataset.

Figure 8

Frequency Chart for Custcat



Note. This chart displays the percentage of custcat in the Telco Extra dataset.

Churn

Customer churn in the Telco Extra dataset is represented as a binary categorical variable. A value of 0 indicates customers who did not churn and a value of 1 indicates customers who churned. As shown in Table 9, 72.6% of customers did not churn, while 27.4% of customers experienced churn. This shows that churn is a relatively infrequent outcome within the sample. This imbalance suggests that most customers remained with the company during the observation period. It is important to note that the dataset does not specify the length of time or observation window used to classify a customer as churned. This information can greatly impact key insights as a churn window of 1 year would differ greatly than a window of 1 month.

Table 9

Frequency Statistics for Customer Churn

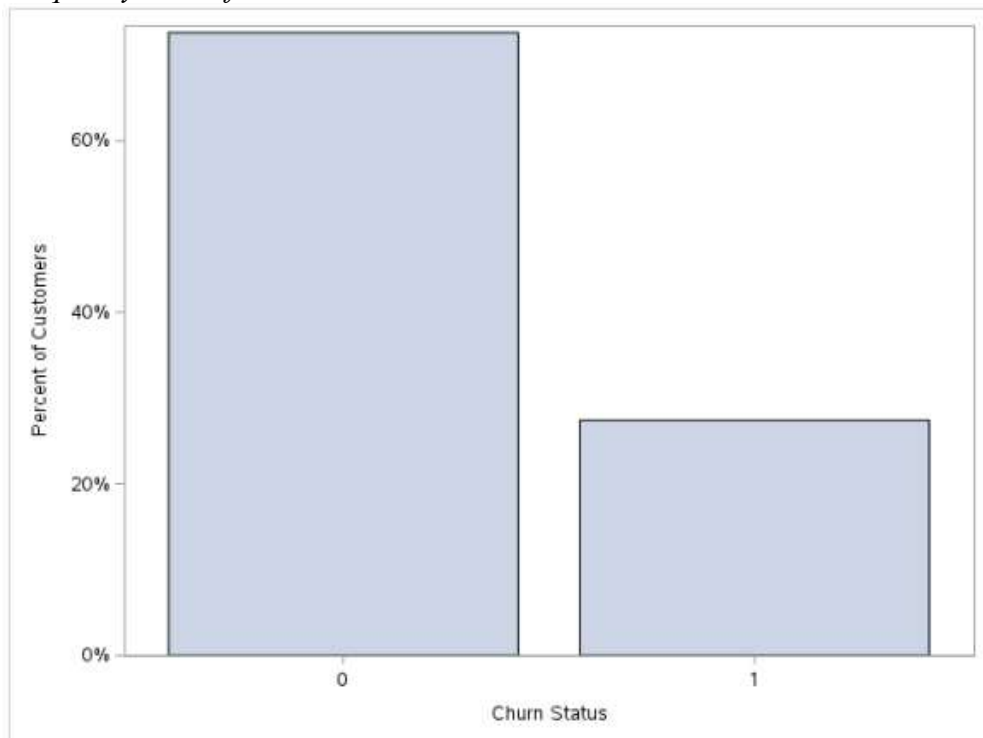
churn				
churn	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	726	72.60	726	72.60
1	274	27.40	1000	100.00

Note. This table summarizes descriptive statistics for customer churn in the Telco Extra dataset. A value of 0 indicates customers who did not churn and a value of 1 indicates customers who have churned.

Figure 9 visually reinforces this pattern by illustrating a substantially higher proportion of non-churned customers compared to churned customers. The observed churn rate offers useful context for later analysis and serves as a reference point for the logistic regression model used to examine factors associated with customer churn.

Figure 9

Frequency Chart for Customer Churn



Note. This chart displays the percentage of customers who did / did not churn in the Telco Extra dataset.

Logistic Regression Analysis (Churn)

The logistic regression model used in this analysis is a binary logit model. This model is appropriate because the dependent variable (churn) has only two possible outcomes. Binary logistic regression is designed for situations in which the response variable represents the presence or absence of an outcome and estimates the probability that a given event will occur based on a set of explanatory variables (Elliott, 2023). In this context, the model predicts the likelihood that a customer will churn using a combination of demographic and service-related predictors. The model is estimated in SAS using Fisher's scoring method, an iterative optimization technique that refines parameter estimates until convergence is achieved. Fisher's scoring is well suited for generalized linear models such as logistic regression because it efficiently updates coefficients based on the expected information matrix. It allows the model to converge reliably even when relationships between predictors and the outcome are nonlinear (Gu et al., 2011). Together, the binary logit framework and Fisher's scoring provide a robust foundation for modeling customer churn and interpreting the relative influence of each explanatory variable.

The response profile confirms that the logistic regression model is estimating the probability of churn = 1. Of the 1,000 customers in the dataset, 726 did not churn and 274 experienced churn, corresponding to a churn rate of 27.4%. This distribution mirrors the descriptive results reported earlier and establishes the baseline outcome frequency for the predictive model.

The model fit statistics, illustrated in Table 10, demonstrate that the full logistic regression model provides a substantially better fit than the intercept-only (null) model. When the explanatory variables are added, the -2 Log Likelihood decreases from 1174.39 to 1045.01,

and the Akaike Information Criterion (AIC) decreases from 1176.39 to 1073.01. These reductions indicate that the inclusion of age, years at current address, gender, education level, marital status, region, and customer category meaningfully improves the model's ability to explain variation in customer churn. In other words, the predictors collectively contribute information beyond what would be expected by chance, supporting the use of the full model for analyzing factors associated with churn.

Table 10
Model Fit Statistics

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1176.394	1073.014
SC	1181.301	1141.722
-2 Log L	1174.394	1045.014

Note. This table compares the intercept-only model with the full logistic regression model. Lower values of =2 Log Likelihood and AIC indicate improved model fit showing that the inclusion of predictors enhances the model's ability to explain customer churn.

The Type 3 Analysis of Effects (Table 11) evaluates the overall contribution of each predictor to the logistic regression model while controlling for all other variables. As shown in Table 11, education level ($x^2 = 18.56$, $p = .0010$), age ($x^2 = 10.30$, $p = .0013$), and years at current address ($x^2 = 13.38$, $p = .0003$) are statistically significant predictors of customer churn at the .05 level. These results indicate that differences in education, customer age, and residential stability are meaningfully associated with the likelihood of churn.

Table 11
Type 3 Analysis of Effects

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
gender	1	0.0631	0.8017
marital	1	1.8599	0.1726
ed	4	18.5640	0.0010
region	2	0.6177	0.7343
custcat	3	11.6013	0.0089
age	1	10.2959	0.0013
address	1	13.3777	0.0003

Note. This table presents the Type 3 Analysis of Effects from the logistic regression model predicting customer churn. Wald chi-square statistics test the overall contribution of each predictor while controlling for all other variables. Effects with p-values less than .05 are considered statistically significant (zyBooks, 2022).

In contrast, gender ($p = .8017$), marital status ($p = .1726$), and region ($p = .7343$) do not demonstrate statistically significant effects, suggesting that these demographic and geographic factors do not independently contribute to explaining churn in this model. Customer category (custcat) approaches significance ($p = .0089$) but does not meet the conventional .05 threshold, indicating a potential but inconclusive relationship with churn.

Table 12 is the odds ratio estimates. They indicate the direction and strength of each predictor's relationship with customer churn while holding other variables constant. Values less than 1.00 reflect reduced odds of churn, whereas values greater than 1.00 indicate increased odds.

Table 12

Odds Ratio Estimates

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
gender 0 vs 1	0.962	0.714	1.297
marital 0 vs 1	1.233	0.912	1.667
ed 1 vs 5	0.355	0.180	0.702
ed 2 vs 5	0.419	0.228	0.770
ed 3 vs 5	0.544	0.296	1.001
ed 4 vs 5	0.831	0.463	1.489
region 1 vs 3	1.083	0.749	1.565
region 2 vs 3	1.157	0.803	1.667
custcat 1 vs 4	0.886	0.586	1.341
custcat 2 vs 4	0.730	0.478	1.116
custcat 3 vs 4	0.478	0.302	0.758
age	0.974	0.958	0.990
address	0.957	0.935	0.980

Note. This table reports odds ratio estimates from the logistic regression model predicting customer churn. Odds ratios less than 1.00 indicate decreased likelihood of churn, whereas values greater than 1.00 indicate increased likelihood. Confidence intervals that include 1.00 denote effects that are not statistically significant.

Both continuous predictors are statistically significant. Age has an odds ratio of 0.974, indicating that each additional year of age reduces the odds of churn by approximately 2.6%. Years at current address has an odds ratio of 0.957, meaning that each additional year at the same residence is associated with about a 4.3% decrease in the likelihood of churn. These results suggest that older and more residentially stable customers are less likely to leave the company.

Several educational categories contrasts are also significant. Compared to the highest education level, customers in education levels 1 and 2 exhibit substantially lower odds of churn (OR = 0.355 and OR = 0.419, respectively). In contrast, gender, marital status, region, and customer category do not demonstrate statistically significant effects, as their confidence intervals include 1.00.

Hypothesis

- **Null Hypothesis (H_0):**

There is no statistically significant relationship between customer churn and any of the explanatory variables (age, years at current address, gender, education level, marital

status, region, and customer category). That is, none of these variables significantly affect the probability that a customer will churn.

- **Alternative Hypothesis (H_1):**

At least one of the explanatory variables (age, years at current address, gender, education level, marital status, region, or customer category) has a statistically significant relationship with customer churn. In other words, one or more of these variables significantly influences the probability that a customer will churn.

Strength and Direction of Relationship

The results indicate that churn is most strongly associated with age, years at current address, and education level. Both age and residential tenure exhibit negative relationships with churn, as shown by odds ratios below 1.00. Each additional year of age ($OR = 0.974$) and each additional year at the same address ($OR = 0.957$) reduces the likelihood of churn. This suggests that older and more residentially stable customers are less likely to leave the company. Although the effect of a single year is modest, these relationships accumulate over time and reflect meaningful differences in customer retention.

Education level also demonstrates a significant relationship with churn. Relative to the highest education category, customers in the lowest education groups exhibit substantially lower odds of churn, indicating that customers with higher levels of education are more likely to leave the company. In contrast, gender, marital status, region, and customer category do not display statistically significant effects, with odds ratios near 1.00 and confidence intervals that include 1.00. Together, these results suggest that churn in this dataset is driven more by factors related to stability and socioeconomic positioning than by basic demographic or geographic characteristics.

Multiple Linear Regression (Income)

To examine the factors associated with customer income, a multiple linear regression model was developed using income as the dependent variable and age, years at current address, gender, education level, marital status, region, and customer category as predictor variables. Unlike the logistic model used for churn, this analysis treats income as a continuous outcome and estimates how each explanatory variable contributes to changes in income while holding all other variables constant. Multiple linear regression is well suited for this purpose as it enables the assessment of both the direction and magnitude of relationships between predictors and a continuous response variable (zyBooks, 2022).

The analysis of variance (ANOVA) results indicate that the multiple linear regression model is statistically significant overall. As shown in Table 13, the model yields an f-statistic of 13.50 with a corresponding p-value of less than .0001. This result demonstrates that the set of predictor variables collectively explains a significant portion of the variation in customer income. This means that the model provides a better fit to the data than a model with no predictors and that at least one of the independent variables contributes meaningfully to predicting income. The ANOVA findings therefore support proceeding with interpretation of the model's fit statistics and individual parameter estimates.

Table 13
ANOVA

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	1729483	133037	13.50	<.0001
Error	986	9717511	9855.48820		
Corrected Total	999	11446995			

Note. This table presents the analysis of variance (ANOVA) for the multiple linear regression model predicting customer income. The F statistic tests whether the set of predictor variables collectively explains a significant portion of variance in income. A p value less than .05 indicates that the model is statistically significant overall (zyBooks, 2022).

These model fit statistics summarize both the strength and usefulness of the regression model. The model yields an r-square of 0.1511 and an adjusted r-square of 0.1399, indicating that approximately 14–15% of the variance in customer income is explained by the combined set of predictors. In multiple regression, r-squared serves as a measure of the strength of the relationship between the dependent variable and the set of independent variables (Elliott, 2023). Although this level of explanatory power is modest, it is consistent with models that rely on demographic and categorical predictors to explain income, a variable influenced by many unobserved factors.

The ANOVA results from Table 13 confirm that the regression equation has predictive value, as the overall model is statistically significant ($p < .0001$). In this context, a low p-value indicates that at least one of the regression coefficients differs from zero and contributes meaningful information to the model (Elliott, 2023). The Root MSE of 99.27 reflects the average prediction error in the units of income and highlights the wide dispersion present in the outcome variable. Together, these statistics show that while the model explains only a portion of income variability, it nonetheless provides statistically significant insight into the factors associated with customer income.

Hypothesis

- **Null Hypothesis (H_0):**

There is no statistically significant relationship between customer income and any of the explanatory variables (age, years at current address, gender, education level, marital status, region, and customer category). In other words, all regression coefficients are equal to zero, and the model has no predictive value for income.

- **Alternative Hypothesis (H_1):**

At least one of the explanatory variables (age, years at current address, gender, education level, marital status, region, or customer category) is significantly related to customer income. That is, one or more predictors has a regression coefficient that differs from zero and contributes to explaining variation in income.

Strength and Direction of Relationship

The multiple linear regression results indicate that customer income is most strongly associated with age and education level, with more limited influence from customer category. Age exhibits a positive and statistically significant relationship with income ($B = 2.84$, $p < .0001$), indicating that income increases as customers grow older. Education level shows a clear directional pattern as customers in lower education categories have significantly lower predicted income relative to the highest education group (e.g., $B = -52.05$ for level 1 and $B = -34.60$ for level 2). These results reinforce the expected positive relationship between education and earnings.

Customer category demonstrates a modest effect with category 1 associated with lower income relative to the reference group ($\beta = -21.02$, $p = .0257$). In contrast, years at current address, gender, marital status, and region do not exhibit statistically significant effects. The strength and direction of the relationships suggest that income in this dataset is driven primarily by age and education, while other demographic and geographic factors contribute little independent explanatory power.

Summary

This analysis examined customer characteristics, churn behavior, and income patterns within the Telco Extra dataset using descriptive statistics, logistic regression, and multiple linear regression. The descriptive analysis established a baseline profile of the customer population, revealing a predominantly working-age customer base, high residential mobility, a highly right-skewed income distribution, and relatively balanced representation across gender, marital status, region, and customer categories. These patterns provided essential context for the predictive models.

The logistic regression model predicting churn identified age, years at current address, and education level as statistically significant predictors. The direction of these relationships indicates that older customers and those with greater residential stability are less likely to churn, while customers in the highest education category exhibit greater likelihood of churn relative to lower education groups. These results support the alternative hypothesis for churn, confirming that at least one explanatory variable significantly influences the probability of customer attrition. The strength of these relationships is modest on a per-unit basis but meaningful in aggregate, particularly for age and residential tenure. In contrast, gender, marital status, region, and customer category did not demonstrate statistically significant effects.

The multiple linear regression model predicting income was also statistically significant, though it explained a modest portion of variance. Age and education levels emerged as the strongest predictors, with income increasing as customers grow older and as education rises. Customer category showed limited influence, while years at current address, gender, marital status, and region were not significant predictors. These findings support the alternative hypothesis for income, demonstrating that at least one predictor contributes meaningfully to explaining income. The direction of effects aligns with expectations, particularly for age and education.

Several findings were unexpected. The absence of a data dictionary limited interpretability for some categorical variables, particularly gender and customer category. The near-uniform distribution across regions was also surprising, as customer bases are often geographically uneven. Additionally, the lack of significance for variables such as gender and marital status in both models suggests that surface-level demographics play a smaller role than stability and socioeconomic factors in predicting both churn and income.

In conclusion, the statistical results support the stated hypotheses and demonstrate that meaningful relationships exist within the dataset. While neither model explains all variation in churn or income, both provide valuable insight into how age, stability, and education shape customer behavior and financial characteristics.

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